

11

[This question paper contains 8 printed pages.]

Your Roll No.....

आपका अनुक्रमांक.....

Sr. No. of Question Paper : 5879

I

Unique Paper Code : 32341502

Name of the Paper : Theory of Computation

Name of the Course : B.Sc (H) Computer Science

Semester : V (Admissions 2019-2021)

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Section-A (Question No. 1) is compulsory.
3. Attempt any four Questions from Section-B (i.e, from Question No. 2 to 7).
4. Parts of a Question must be answered together.
5. Consider $\Sigma = \{a, b\}$ for all the questions unless specified otherwise.

P.T.O.

Section - A

1. a) Given the language $L = \{ab, aa, baa\}$, determine whether the string 'abaabaaabaa' is in L^* ? (2)
- b) Consider a language consisting of all the words that end with sub-string 'bb'. Write a regular expression for the language. (3)
- c) Let us consider the regular expression $(a+b)^*a(a+b)^*b(a+b)^*$. Show that this is equivalent to $(a+b)^*ab(a+b)^*$. (3)
- d) A context-free grammar (CFG) G is given by the following productions (3)

$$S \rightarrow aS \mid A$$

$$A \rightarrow aAb \mid bAa \mid A$$

Determine whether the string 'abababb' can be generated by the grammar given above.

e) Consider the following CFG: (5)

$$S \rightarrow aB \mid bA$$

$$A \rightarrow a \mid aS \mid bAA$$

$$B \rightarrow b \mid bs \mid aBB$$

i) Describe the language generated by the given grammar.

ii) Is the language ambiguous or unambiguous? Justify your answer.

f) Using pumping lemma, prove that the language a^2nb^n is not regular. (4)

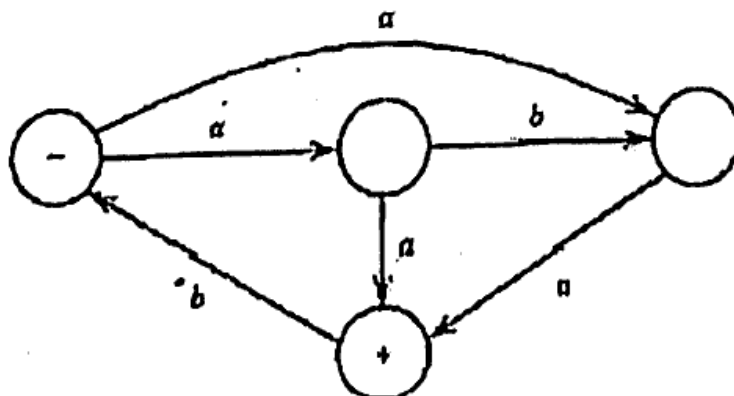
P.T.O.

g) Design a Turing machine for the language $a^n b^n a^n$. (5)

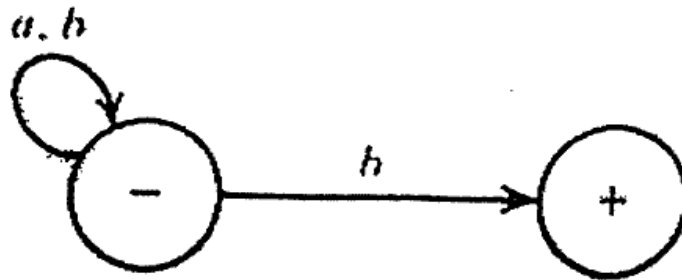
Section - B

2. a) Consider the language $S = \{ aa, ba, ab, bb \}$.
Describe the language S^* . (4)

b) Convert the following Non-deterministic finite automata(NFA) into a regular expression. Also, construct a automata(DFA) for the given NFA. (6)



3. a) Consider the following transition graph(TG). Describe the language accepted by it. Also construct a DFA for the given TG. (6)



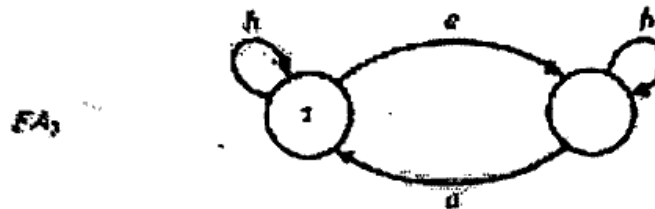
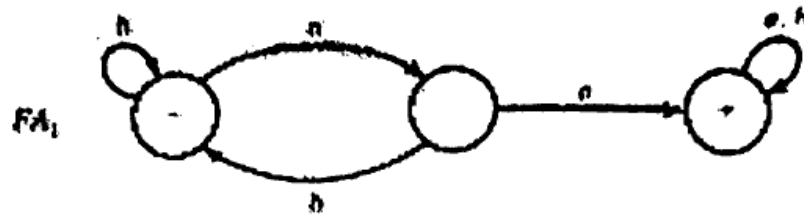
- b) Construct a push down automata(PDA) for the language L given below: (4)

$$L = \{a^n b b^n \mid n > 0\}$$

4. a) Construct a CFG for the language containing all words having different first and last letters.(4)
- b) Design a Turing Machine which computes the function $f(w) = ww$. (6)

P.T.O.

5. a) Find the union machine for the given deterministic finite automata FA1 and FA2: (6)



- b) Design a context-free grammar for the the language described by regular expression a^*b^* .

(4)

6. a) Build a deterministic finite automaton for the language of all words that do not end with ba.(4)

b) Consider the following context-free grammar G(6)

$$S \rightarrow SS|ab|ba|A$$

i) Which of the following words can be generated by the given grammar: ababba, ababbba

ii) Draw derivation tree for the word that belongs to the language produced by the grammar.

iii) Is the language ambiguous?

7. a) Consider the following CFG. Convert it into Chomsky Normal Form. (5)

$$S \rightarrow aB \mid bA,$$

$$A \rightarrow a \mid aS \mid bAA,$$

$$B \rightarrow b \mid bs \mid aBB$$

P.T.O.

- b) Prove that the language $\{a^n b^n c^n d^n \text{ for } n = 1, 2, 3, 4, \dots\}$ is not context-free. (5)